

Fractional Knapsack Problem



Ek Beginner-Friendly Guide, tumhare style mein!

Context Link: Jaise tum *brass lights* ka business handle karte ho, aur tumhe limited space wale box mein maximum profit wali lights pack karni ho, bas wahi logic yahan lagta hai. Ya phir socho tum gym jate ho, aur ek chhota shaker hai jisme tumhe *high-protein items* (jaise *whey*, *soyabean*) pack karne hain jinki nutritional value sabse jyada ho aur weight kam! Zomato gold se pizza order karne jaisa greedy approach, jahan max taste kam paiso mein chahiye! 😊

1. Kya hai yeh Fractional Knapsack? 🤔

Knapsack ka matlab hota hai **Bori (Bag)**. Imagine karo ek chor (thief) ek kirane ki dukan (grocery store) mein gaya hai. Uske paas ek bori hai jiski ek maximum capacity hai (maan lo 20 kg). Dukan mein alag-alag items hain (jaise dry fruits, daal, chawal) jinke apne-apne weights aur total prices hain.

Chor ka Target (Greedy Approach): Usko apni bori mein aise items bharne hain jisse uska total profit MAXIMUM ho. Agar bori bharne wali ho aur koi item pura nahi aa raha, to wo usey kaat kar ya thoda hissa (fraction) le ja sakta hai (kyunki ye kirana store hai, shakkra/dry-fruits ko openly weight karke liya ja sakta hai).



Visualizing The Knapsack

Bag Capacity: 20 KG

📦 Item 1: Dry Fruits (Weight: 5kg, Price: ₹40) ➡ Per kg: ₹8

📦 Item 2: Daal (Weight: 4kg, Price: ₹24) ➡ Per kg: ₹6

📦 Item 3: Chawal (Weight: 6kg, Price: ₹30) ➡ Per kg: ₹5

Chor ka greedy dimaag: "Sabse mehenga per kg wala pehle uthao!"

2. Step-by-Step Logic (Algorithm) 🚶

Is problem ko solve karne ke liye humara **Greedy Approach** kya hona chahiye?

Step 1: Per KG Price nikalo 📊

Har ek item ka **Per_Kg_Price = Total_Price / Total_Weight** calculate karo.

Step 2: Items ko Sort karo

Items ko unke Per KG Price ke hisaab se **Decreasing Order** (sabse mehenga pehle) mein sort karo.

Step 3: Bori (Knapsack) mein bharo

Sabse mehenge item se shuru karo:

- Agar bag mein poora item aa sakta hai ($\text{Weight} \leq \text{Bag Capacity}$), to poora daal do. Bag ki capacity minus karo, aur profit add karo.
- Agar bag mein poora item NAHI aa sakta, to jitni jagah bachi hai, utna hi (fraction) bharo aur bag ko poora bhar do.

3. Ek Zabardast Example se Samjhein!

Maan lo dukan mein ye 4 items hain aur chor ki bori ki **Capacity = 20 kg** hai.

Item	Total Price (₹)	Total Weight (kg)	Per KG Price (₹)
Item 0	40	5	$40/5 = 8$
Item 1	24	4	$24/4 = 6$
Item 2	30	6	$30/6 = 5$
Item 3	21	7	$21/7 = 3$

Items already sorted hain decreasing order mein. Chalo bori bharte hain:

- **Pick Item 0 (5kg, ₹40):** Bag mein jagah hai 20kg. Poora daal do!
New Capacity: $20 - 5 = 15\text{kg}$. *Profit:* ₹40.
- **Pick Item 1 (4kg, ₹24):** Bag mein jagah hai 15kg. Poora daal do!
New Capacity: $15 - 4 = 11\text{kg}$. *Profit:* $40 + 24 = ₹64$.
- **Pick Item 2 (6kg, ₹30):** Bag mein jagah hai 11kg. Poora daal do!
New Capacity: $11 - 6 = 5\text{kg}$. *Profit:* $64 + 30 = ₹94$.
- **Pick Item 3 (7kg, ₹21):** Bag mein jagah hai bas 5kg, par item hai 7kg ka. 
Fractional Time! Hum sirf 5kg uthayenge.
Cost of 5kg = $5 * \text{Per KG price (₹3)} = ₹15$.
New Capacity: 0kg. *Total Profit:* $94 + 15 = ₹109$.

4. Python Code (Bilkul tumhare PyCharm style mein) 🐍💻

Ab dekho iska asaan sa Python code. Python mein triplets banana aur sort karna bahut easy hai.

```
def fractional_knapsack(weights, prices, capacity):
    n = len(weights)

    # Step 1: Create a list of items with [price, weight, per_kg_price]
    items = []
    for i in range(n):
        per_kg_price = prices[i] / weights[i]
        items.append([prices[i], weights[i], per_kg_price])

    # Step 2: Sort items based on per_kg_price in descending order
    # Bubble sort logic for easy understanding
    for i in range(n):
        for j in range(i + 1, n):
            if items[i][2] < items[j][2]:
                # Swap items
                items[i], items[j] = items[j], items[i]

    # Step 3: Fill the knapsack
    total_profit = 0

    for item in items:
        price = item[0]
        weight = item[1]
        per_kg_price = item[2]

        if capacity >= weight:
            # Pura item bag mein aa jayega
            capacity -= weight
            total_profit += price
        else:
            # Bag bharne wala hai, fraction (tukda) uthao
            total_profit += (per_kg_price * capacity)
            capacity = 0 # Bag full ho gaya!
            break # Ab aur item nahi aa sakta

    return total_profit

# Let's test it!
weights = [5, 4, 6, 7]
prices = [40, 24, 30, 21]
capacity = 20
```

```
max_profit = fractional_knapsack(weights, prices, capacity)
print("Maximum Profit is:", max_profit)
```

5. Important Interview Points

- **Greedy Algorithm:** Yeh problem Greedy approach se easily solve ho jati hai kyunki har step pe hum "Sabse jyada fayda (Per KG Price)" dekhte hain.
- **Fractional vs 0/1 Knapsack:**
 - *Fractional:* Item ka tukda (fraction) le ja sakte hain (jaise chawal, protein powder). Greedy approach works!
 - *0/1 Knapsack:* Ya toh pura item lo ya bilkul mat lo (jaise TV, Fridge, ya Decorative Brass Light). Tum aadhi light tod kar nahi le ja sakte! Iske liye hume **Dynamic Programming** use karni padti hai.
- **Real life application:** Jaise job scheduling! Maan lo tumhare paas limited time hai aur tum alg-alg freelance Python/Data Analyst gigs karna chahte ho. Tum wo gig pehle uthaoge jisme hourly rate sabse high ho (Per Hour Price = Per Kg Price).

Pro Tip for Soubhik: Jaise tum Naukri/LinkedIn par job hunt karte ho ATS score badhane ke liye, wese hi Interviews mein Greedy vs DP ka concept clear hona tumhe ATS se lekar Technical round tak aage rakhega! Keep coding! 